# 4. Intro to Graph Theory

At this point in your mathematical careers, you probably know a lot about graphs of functions. Over the next few weeks, you will all learn more about a new kind of graph.

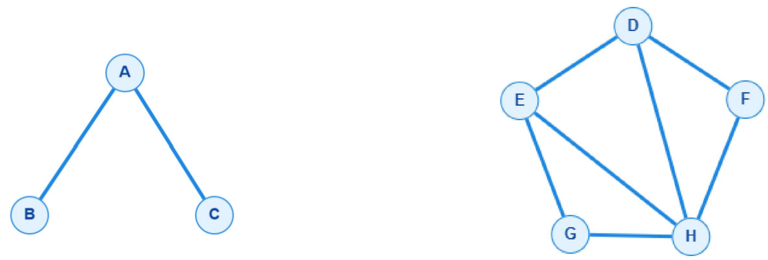



Figure 1. Two examples of graphs.

In the subfield of mathematics known as *graph theory*, a **graph** is a mathematical object that is built out of vertices (represented by dots) that are connected by edges (represented by lines "connecting" the dots). Put more rigorously:

Definition. A (finite) **simple graph** $G$ consists of the following data:

- a set $V(G) = \{v_1, v_2, \dots, v_n\}$ called the **vertex set** of $G$ and
- a set $E(G) = \{\{v_i, v_j\} \mid \text{where } i \neq j \text{ and there exists an edge between } v_i \text{ and } v_j\}$ called the **edge set** of $G$.

A graph $G$ is best represented with a picture. For each vertex $v \in V(G)$, we place a dot/bullet/node in the plane. For each edge $e \in E(G)$ between two vertices $v$ and $w$ in $V(G)$, we draw a line connecting the dots that represent $v$ and $w$. See Figure 1.

**Remark.** *Some things to keep in mind regarding the second bullet point in the definition of a graph:*

- *Furthermore, note that the elements of $E(G)$ are themselves sets, and so we make no distinction between $\{v_i, v_j\}$ and $\{v_j, v_i\}$; they represent the same edge.*
- *In practice, the notation $\{v_i, v_j\}$ can get old very fast, so we will often shorthand the edge labelings to $v_iv_j$, and you may even find it convenient to just label them with something else (like a number).*
- *Finally, notice that our definition of a graph does not allow for "repeated edges." That is, we do not allow for there to be more than one distinct edges between the same pair of points. One can modify the definition of a graph to allow for repeated edges, but many authors (including us) call this a **multigraph**. In particular, a multigraph* is not *a simple graph.*

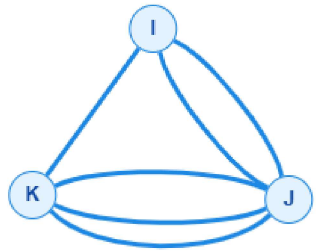



FIGURE 2. A graph with repeated edges (a multigraph).

**The only graphs we will care about this summer are simple graphs. Since all our graphs will be simple, we will often drop the adjective.**

**Exercise 25.** *Call the two graphs in Figure 1 $X$ (the one on the left) and $Y$ (the one on the right).*

1. *What are $V(X)$ and $V(Y)$?*
2. *What are $E(X)$ and $E(Y)$?*
3. *Is $X$ a simple graph?*
4. *Is $Y$ a simple graph?*
5. *Make a new graph $Z$ by taking the union of $X$ and $Y$. That is, put $V(Z) = V(X) \cup V(Y)$ and $E(X) \cup E(Y)$. This is an example of a* ***disconnected graph****. It is probably kind of obvious from a picture whether or not a graph is connected or not, but can you come up with a* rigorous *definition of a* ***connected graph****?*

**Exercise 26.** *There are two distinct (simple) graphs $G$ for which $|V(G)| = 2$. Draw them, describe the edge and vertex sets.* [13] *What about the unique graph with $|V(G)| = 1$? $|V(G)| = 0$?*

DEFINITION. Let $G$ and $H$ be graphs. We say that $H$ is a **subgraph** of $G$ if and only if (possibly after renaming vertices/edges) we have the containment of vertex sets $V(H) \subseteq V(G)$ and the containment of edge sets $E(H) \subseteq E(G)$.

[13] Hint: We never said that $E(G)$ had to be a *nonempty* set.

**Example.** *Assume all the graphs below are connected. The graph to the bottom left is called* $K_5$*, and the three graphs next to it are examples of subgraphs. It may be hard to see that the last two are subgraphs. To see what we claim is true, consider the definition that follows.*

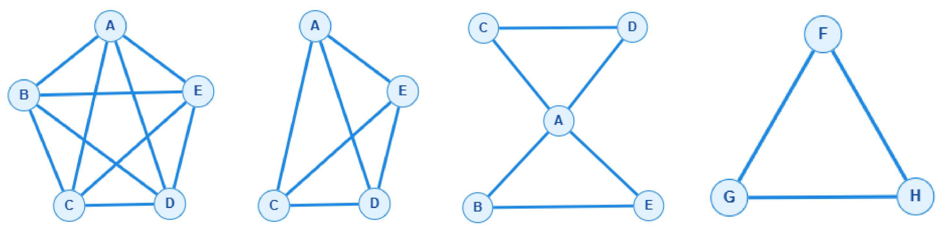



FIGURE 3. The leftmost graph is called $K_5$, and 3 of it's subgraphs are to the right.

DEFINITION. Let $G$ and $H$ be graphs. We say that $G$ and $H$ are **equivalent** graphs if and only if (possibly after relabeling the vertices/edges), we have equalities $V(G) = V(H)$ and $E(G) = E(H)$.

So the equality of graphs really comes down to the equality of sets, and we may even have to relabel elements of the sets. This explains why two representations of the same graph might pictures may *appear* different.

**Exercise 27.** *Convince yourself that Figure 3 really depicts three subgraphs of* $K_5$*. Are they all distinct subgraphs? How do you know?*

**Exercise 28.** *Are the following two graphs in Figure 4 equivalent? How do you know?*

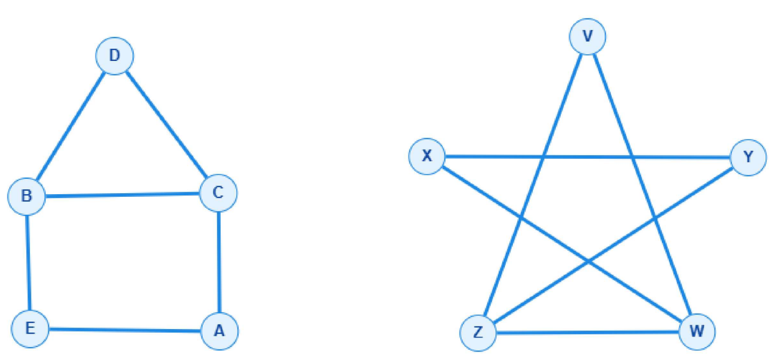
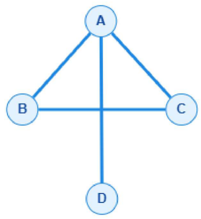



FIGURE 4. Two(?) graphs.

**Exercise 29.** *Assume all graphs are connected. Group the letters of equivalent graphs together.*

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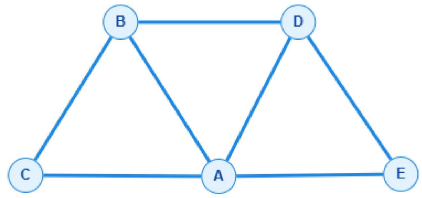


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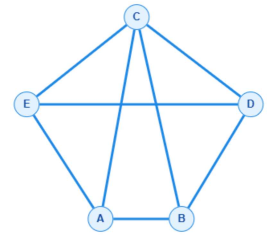


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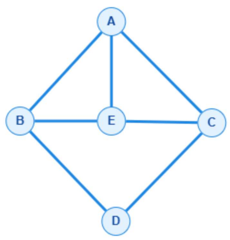


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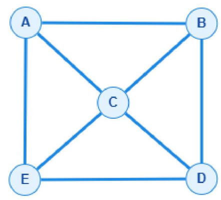


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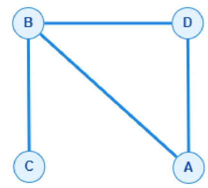


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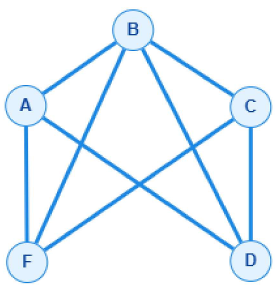


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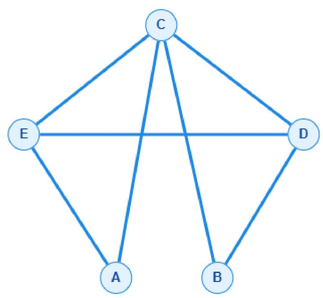


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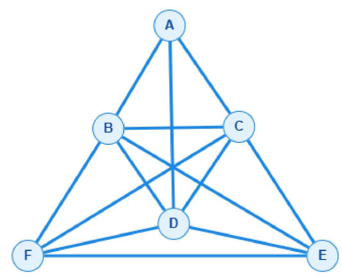


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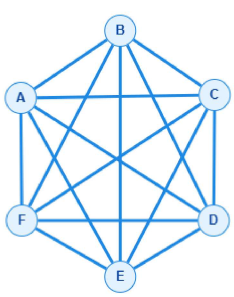


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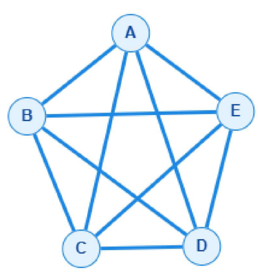


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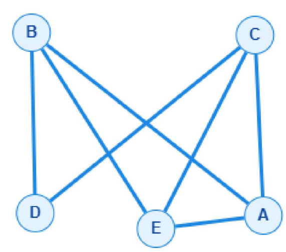


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